

Course report
SF2321 Applied Statistical Analysis
Fall 2024
Maria Solevid

Introduction

The course SF2321 Applied Statistical Analysis is a second-cycle method course for MA students in Political Science (MAPS), Political Science: Environmental Governance and Behavior (MAPSE), European Studies (MAES), International Administration and Global Governance (IAGG), Political Communication (POLCOM), Criminology¹ and Sociology.² The course underwent significant revisions in both learning outcomes and content in 2023³. Thus, 2024 was the second time the course was held in its current form. The course teaches research design and statistical techniques for causal inference.

Four teachers, all associate professors, from the Department of Political Science taught on the course in 2024 (same as in 2023): Mattias Agerberg, Rasmus Broms, Maria Solevid and Aksel Sundström. All teachers were involved in supervision and examination of the method theses.

At the start of the course 2024, 42 students were registered on the course, 37 first-time registered and 5 re-registered⁴. Two students changed course after about one week and one student paused the course after a couple of weeks which means that the course in the end had 34 first-time registered and 4 re-registered students. All of the remaining 34 first-time registered were active and took credits. Of the 34 first-time registered students, 2 were from IAGG, 8 from MAES, 12 from MAPS, 3 from MAPSE, 1 from POLCOM, 6 from Sociology and 2 free-standing students.

The course was taught on campus with the addition of some well-thought through digital teaching events. Based on previous student cohorts positive feedback we have deliberately kept the pre-recorded introduction to the Stata tutorials, where the instructor demonstrates the code and procedures in Stata relevant for the assignment (1-6). We also carry out three advice sessions related to the Method thesis on Zoom, this as these sessions take place just before Christmas and in early January. Since we demand almost weekly assignments from the students for the first 9 credits, we grade these assignments quick, often only within a couple of days.

The 2024 course content was almost identical to the 2023 course. We improved the recap and introduction to regression workshops, the latter also meant that the responsible teacher recorded new tutorial videos. We reviewed lecture slides and fine-tuned assignment instructions for a couple of assignments. We also created an AI policy to clarify in what instances generative AI is OK and not OK to use.

¹ MA students in Criminology participate in one of the two quantitative method courses every other year (2019, 2021 and so on).

² The course started in 2007 and was then a formal collaboration between Department of Journalism, Media and Communication, Department of Sociology and Work Science and Department of Political Science (coordinating department), but the participating MA programs has shifted over the years. Throughout the years, the course has been revised several times, and big revisions took place in 2012 and 2018.

³ The course revisions are described in the course report for the 2023 course edition

<https://canvas.gu.se/courses/81316/gradebook#>⁴ Two additional students re-registered during the course. Also, three students who started the course changed to SF2324 after two weeks.

Course structure

Table 1 below summarizes the course structure in 2024.

Table 1 Overview of course structure, 2024

	Course structure	Responsible teacher
Week 1-2	Introduction and recap Assignment 1: Design for causal inference (1 hec)	Group assignment, oral presentation at compulsory seminar Maria Solevid/Aksel Sundström (intro/recap) Mattias Agerberg (A1)
Week 2-3	Assignment 2: Introduction to linear regression (2,5 hec)	Group assignment, paper Aksel Sundström
Week 4	Assignment 3: Heterogenous effects in linear regression (1,5 hec)	Group assignment, paper Maria Solevid
Week 5	Assignment 4: Panel regression (1 hec)	Group assignment, paper Rasmus Broms
Week 6	Assignment 5: Difference-in-difference (1,5 hec)	Group assignment, paper Rasmus Broms
Week 7	Assignment 6: Synthetic controls (1,5 hec)	Group assignment, paper Mattias Agerberg
Week 8-10	Method thesis (6 hec)	Individual assignment, paper and oral presentation at compulsory seminar Maria Solevid (all teachers involved)

The course starts with a course introduction and a recap lecture on statistical inference and a recap lecture on basic statistics (recap lecture on the variable language, the data matrix, variable/variable units, descriptive statistics, cross-tabs, correlations and a first glance at linear regression). After the recap follows the lectures related to assignment 1: designs for causal inference. This smaller assignment is examined at a mandatory seminar where the students shortly presents a research design. The lectures and assignment aim introducing and training students how to set up causal research questions, how to formalize the expectations and the counterfactuals using the potential outcomes framework. The potential outcomes framework returns in all lectures and assignments throughout the course.

After the first assignment, we provide a tutorial that introduces the software Stata. Once this is finished, we move on to linear regression (bivariate and multiple regression) and the different parameters estimated, dummy variables and a shorter introduction to regression diagnostics. We also introduce the Unexpected event during survey design (UESD), a quasi-experimental design using cross-sectional survey data to reach causal conclusions. We continue with linear regression during the workshop on heterogenous effects in linear regression in which we teach how to think, use, interpret and understand regression models with an interaction term. UESD comes back side by side with many other examples.

After these $\approx 2,5$ weeks focusing on linear regression, the course moves on to applied methods for causal inference: panel regression, difference-in-difference and synthetic controls. The panel regression workshop introduces how to handle repeated observations over time and the difference between pooled OLS, fixed effects and first-difference models. The difference-in-difference workshop introduces the design for how to compare trends in two groups (countries, municipalities, individuals) when one of the groups is exposed to a treatment (a one-time treatment or a staggered treatment), for example a reform, and how to use regression models to test this empirically. Finally, the synthetic control method introduces how one can design and test the causal effect of a specific event on a single unit (such as the German

unification, or the effect of a new tax in one US state) by creating and comparing with matching cases that represents a synthetic version of the case that was not treated (for example, other European countries, or other US states). The course ends with an individual assignment, the method thesis.

Completion rate

The completion rate after the end of the course was 71 percent among all registered students. The corresponding completion rate was 88 percent among the active first-time registered students (81% in 2023) and 79 percent among all registered students excluding those who changed course or took a study break (78% in 2023). The reason for the increased completion rate among the active first-time registered students is that 31 of 34 of them submitted the final assignment which is a new record.

Of the 30 students who had completed the course during the original submission rounds, 15 were awarded Pass with distinction and 15 Pass (2023: 8 PWD, 16 P). The share of Pass with distinction is 50 percent which is considerably higher than 2023 (33 percent).

Students' assessment of core course elements

Of the 37 registered students, 20 answered the course evaluation survey which equals a response rate of 54,5 percent (2023: 65,6%). The response rate fluctuates over the years, and the 2024 edition show neither the lowest nor highest response rate.⁵ If we consider that 34 students have been active throughout the course, a more accurate response rate is 58 percent. A reminder to fill out course evaluation was stated in the seminar schedule, and students were also reminded at the end of the seminars. Invitation and reminders to the survey were as usual sent out through the SUNET survey system, and one additional reminder was made by the course coordinator through Canvas messages.

Table 2 summarizes the students' assessment of the course in general, the lectures, teachers, computer labs and overall course organization/administration. Most tables present both percentages and means and were applicable, we also include the aggregated average from 2018–2022 for comparison. Please keep in mind that the 2024 and 2023 course significantly differs from the 2018-2022 course edition.

⁵ RR 2023: 65,6 percent of all/58 percent among active (reminder about course evaluation was included in the seminar schedule, and students were also reminded by the seminar chairs at the end of the seminar)
RR 2022: 53,7 percent of all/59,5 percent among active (course evaluation scheduled in seminar program)
RR 2021: 46 percent of all/61 percent among active (verbal reminder during Zoom seminar to fill out the course evaluation survey after the seminar)
RR 2020: 53 percent among all/76 percent among active (verbal reminder during Zoom seminar to fill out the course evaluation survey after the seminar)
RR 2019: 70 percent among all (scheduled time to fill out course evaluation in classroom)
RR 2018: 59 percent among all (verbal reminder at the end of the seminar to fill out the course evaluation survey)

Table 2 General assessments of the course, lectures, tutorials, teachers and course administration (percent and mean)

	The course in general	Lectures	Stata tutorials	Teachers	Organization and administration of the course
Very good	55	40	50	65	65
Rather good	40	60	40	35	35
Neither bad nor good	5	0	10	0	0
Rather bad	0	0	0	0	0
Very bad	0	0	0	0	0
Total	100	100	100	100	100
N	20	20	20	20	20
Mean 2024	4,5	4,4	4,4	4,7	4,7
Mean 2023	4,6	4,3	4,0	4,5	4,4
Mean 2018-2022	4,2	3,9	4,2	4,3	4,5

Comment: Question formulations: "Overall, how would you assess the....?". Very bad coded as 1 and Very good coded as 5.

Overall, the core course elements receive very positive assessments. 55 percent responds that the course was very good (2023: 67%) and 40 percent (2023: 24%) responded 'rather good'. This means that 95 percent provides a positive assessment (very or rather good). The average on a 1-5 scale, where 1 is very bad and 5 very good, is 4,5 which is on par with 2023 (4,6). 100 percent (2023: 95%) respond that the lectures are rather or very good (mean 4,4), 90 percent (2023: 81%) shares the same view of the Stata tutorials (mean 4,4) and 100 percent (2023: 95%) make positive assessments of the teachers (mean 4,7). 100% (2023: 86%) assesses the organization and administration of the course as rather or very good (mean 4,7). Overall, the 2024 average assessments are very similar or even slightly better compared to the 2023 course. We are also very pleased to see that half of students or more respond "Very good" when it comes to the course in general, teachers, Stata tutorials and course organization, and that none of core course elements are assessed as bad.

Students' assessment of the different workshops

Table 3 summarizes students' assessments of the different workshops throughout the course, and the average assessment is reported both for 2024 and 2023.

Table 3 Evaluations of the different workshops (percent and mean)

	Very good	Rather good	Neither good nor bad	Rather bad	Very bad	Total	N	Mean 2024	Mean 2023
Design for causal inference (A1)	55	40	5	0	0	100	20	4,5	4,6
Recap and introduction to Stata	45	35	20	0	0	100	20	4,3	3,9
Introduction to linear regression (A2)	55	40	5	0	0	100	20	4,5	3,5
Heterogenous effects in linear regression (A3)	65	35	0	0	0	100	20	4,7	4,6
Panel regression (A4)	55	35	10	0	0	100	20	4,5	4,0
Difference-in-difference (A5)	60	35	5	0	0	100	20	4,6	4,2
Synthetic controls (A6)	70	30	0	0	0	100	20	4,7	4,4
Method thesis (A7)	50	40	10	0	0	100	20	4,4	4,1

Comment: Question formulations: "What is your assessment of the different workshops specified below?" Very bad coded as 1 and Very good coded as 5. Recap and introduction to Stata is the only workshop not associated with an assignment.

In general, the average assessments are similar or more positive in 2024 compared to 2023. All workshops receive positive evaluations with average assessment ranging from 4,3 to 4,7. The share of students responding that the workshop is very good ranges between 45 and 70 percent, and the share of positive assessments (very+rather good) ranges between 80 and 100 percent.

We are very happy to see that the workshops *Recap and Stata intro* (no assignment) and *Introduction to regression* respectively, that had average assessments below 4 in 2023, now scores well-above 4. It means that work we put into improving these workshops resulted in the anticipated increase in quality.

Students' assessments of examination forms, literature, and amount of work

As shown in table 4, the examination forms – the six workshop assignments and the method thesis– receives average assessments of 4,4 and 4,3 respectively. 95 and 85 percent of the students respond that these examination forms are rather good or very good. The assessments of the examination forms are very similar to 2023.

Table 4 Evaluation of examination forms

	Workshops assignments as examination forms	Method thesis as examination form
Very good	55	40
Rather good	40	45
Neither bad nor good	5	15
Rather bad	0	0
Very bad	0	0
Total	100	100
N	20	20
Mean 2024	4,5	4,3
Mean 2023	4,4	4,3
Mean 2018-2022	4,1	4,2

Comment: Question formulation: Overall, how would you assess the examinations? Very bad coded as 1 and Very good coded as 5

The course literature consists of one textbook: *Mastering Metrics* by Joshua Angrist and Jörn-Steffen Pischke and several journal articles. In short, the book receives dispersed assessments and an average assessment between Neither good nor bad and Rather good (3,6) (see table 5). The journal articles are on average slightly more appreciated among the students (3,8). In the free-text answers, two responses concern the lack of a textbook where the methods are implemented in Stata. In 2023, as well between 2018-2022, we used the book *Applied Statistics Using Stata*, but we decided to remove the book since it neither used the causal inference language nor covered difference-in-difference and synthetic controls. However, the students get do files and videos from us and are further advised to use google and ChatGPT, but clearly this is not enough. Given that Mastering Metrics is not reviewed as helpful enough, and that some students miss the Stata implementation, we will search for a supplement to Mastering metrics.

Table 5 Evaluation of the literature

	Angrist & Pischke Mastering Metrics	Articles (general assessment)
Very good	25	15
Rather good	25	65
Neither bad nor good	30	30
Rather bad	20	5
Very bad	0	0
Total	100	100
N	20	20
Mean 2024	3,6	3,8
Mean 2023	3,8	-

Comment: Question formulation: "Overall, how would you assess the literature?" followed by the two author names and titles of the two course books. Very bad coded as 1, Very good coded as 5. In addition to the response options displayed in the table, the option "I didn't read this book" was also available but has been excluded from the mean calculations.

The assessment regarding amount of work and how many hours the student spent on course work (inside and outside the classroom) is reported in table 6.

Table 6 Evaluation of amount of work

Amount of work		How much time spent on course work	
Too demanding	5	More than 40 hours	20
Somewhat too demanding	25	31-40 hours	35
About right	70	21-30 hours	25
Not that demanding	0	11-20 hours	20
Not at all demanding	0	Less than 10 hours	0
Total	100	Total	100
N	20	N	20
Mean 2024	3,4	Mean 2024	3,6
Mean 2023	4,2	Mean 2023	4,2
Mean 2018-2022	3,7	Mean 2018-2022	3,9

Comment: "What is your assessment of the amount of work you have been requested to carry out during the course?". Not at all demanding coded as 1, Too demanding coded as 5 "On average, approximately how much time did you spend on coursework, both inside and outside the classroom, per week?". Less than 10 hours coded as 1, More than 40 hours coded as 5.

The results most notably show that 70 percent respond the workload was about right (14 percent in 2023). 30 percent assess the course work as somewhat or too demanding (85 percent in 2023). The big difference between the two years, where perceptions of workload are clearly lower in 2024, are likely a consequence scheduling. The aim is always to have as many submissions as possible on Fridays and start a new workshop the following Monday. In 2023, this was not technically possible to achieve during the first half of the course and resulted, we think, in this high degree of students responding the course was too demanding. In 2024, we were able to once again schedule 6 of 7 assignment deadlines on Fridays. That more than two-thirds of the students assess the workload as about right in 2024, without any changes in the assignments, is telling of how important the schedule logistics is. Regarding how much time the students spent on course work, we also see a decrease between the two years. 20 percent responded they spent more than 40 hours per week on the course work (2023: 38 percent), 35 percent answers 31-40 hours (2023: 43) and 25 percent 21-30 hours (2023: 19 percent). In 2024, 20 percent report having only spent 11-20 hours on the course, the corresponding share in 2023 was 0 percent. In sum, most of the students have not spent more than 40 hours on course work. Judging from the displayed averages, the course was

viewed as considerably less demanding compared to 2023, although the actual course content was almost identical.

Students' learning and expectations

84 percent of the students answer that they experienced a Eureka moment during the course (2023: 67 percent). Among the free-text answers, causality and selection bias, the potential outcomes framework, the synthetic control method, how to interpret heterogenous effects, how to set up and handle causal study designs are examples of mentions of a Eureka moment.

Reflections

From our teacher perspective, we are overall very satisfied with how the course went and we are very impressed with the quality of the students' work, which not least is evident considering that nearly 50 percent of the students received Pass with distinction. We are also very happy that the course is well-received among the students. We the teachers have continued to work closely together to prepare the course and to communicate with each other during the course. In doing this, we could remedy issues that were raised in the 2023 course evaluation and in the course review made by two other teachers in the department. The close contact during the course also means we quickly can remedy minor issues that arises. The collaboration with the education administrator has been excellent.

We are especially pleased to see that the efforts to improve the Recap and the Introduction to Stata led to increased quality and better assessments from the students. Other changes we made was that the abstract submission for the Method thesis was pushed forward two weeks compared to 2023, this to enable that all methods had been introduced and to get more students to choose the difference-in-difference or synthetic control methods. As the share of students who used these methods also increased compared to 2023, we think it was the right move.

However, there is still room for improvement. From the free-text answers, we have identified the following categories of feedback: *Things students wish were covered* and *Things students wish were done different*. In addition to these two broad categories, we also collect responses related to the *Literature* and the *Method thesis* in own categories. Finally, there is one category labelled as *Miscellaneous*. Before going through the free-text answers, we want to thank the students for writing such constructive feedback.

Among things *students wish were covered*, we see responses concerning how to handle missing/bad data, that there should be some recap in the beginning of what a nice table looks like, that we should do more on non-dichotomous treatments and that we should go more in-depth in different methods, and cover survey design and coding with more advanced tool. Additional responses are that we should explain better what no selection bias means in social science and that there should be clearer instructions about causal claims, for example how to use it with cross-sectional data or when it is suitable to have causal claims or not. Finally, one response concerns that we should include a couple of critical articles on statistical methods. One response more specifically asks for more step-by-step STATA guides. This includes what each method is best suited for and for what type of data, what the assumptions for each method are and how to practically do analysis with Stata.

Our aim is to improve how to handle missing data, to recap how to produce good tables, to further strengthen how to think about no selection bias and causality, and non-dichotomous treatments. In the current course structure, we unfortunately don't have room for more in-

depth of different methods, survey design, and advanced coding even though we would very much like too, and we don't think critical articles on statistical methods in general is beneficial to the course. The responses regarding Stata guides is something we will consider, not least in the light of that students raise some criticisms against the main textbook (see below).

Among the things *students wish were done different*, the responses concerns that the intro to OLS could have been shorter as we should expect students to know this and that there is too much focus on causal inference and that time instead could be spent on other regression techniques common regression techniques in the social sciences. One response also state there should be more individual assignments to prepare better for the method thesis. One other response concerns that the DID lecture should be expanded to include how to interpret the results from Stata. Finally, one response concerns that the first tutorial, Introduction to Stata, suffer from differences in students' basic computer knowledge leading to that it was too slow for some students. It is suggested to open up for different speeds and that students can leave earlier if they have finished their tasks. While we unfortunately lack resources to introduce more individual assignments, we can even more remind and push the students to take responsibility for their own learning (this is however already part of the course introduction). The time spent on OLS intro will be reviewed and assessed, and more on interpretations of DID results will be included next year. The issue with the pace of the tutorial that introduces Stata is probably harder to accommodate, but we will nevertheless review it.

The responses about the *literature* align with each other: students who raise feedback about the course literature think Mastering metrics do a bad job teaching the methods. An illustrative quote from one student is "I feel almost like the chapter on difference-in-differences taught me more facts about the banks in the American south during the great depression than about the Difference-in-differences method". The students who criticize Mastering Metrics also suggests other textbooks such as Mixtape and Data Analysis for Social Science (although the latter book has R implementation). We will review and assess if a change of textbook would be more beneficial. There is also one response about that few students knew Stata from before and that a book (or ebook) explaining the syntax would have been helpful. As discussed earlier, we used to have such book but we removed it since it didn't match with the causal inference design or the methods taught.

A number of responses concern issues related to the *Method thesis*. Two responses concern that the (high) expectations on the thesis doesn't square with the word limit (4000 words, excluding references and appendix) which made the assignment more difficult. A suggestion raised is to remove the literature review (the demands on literature review is however already much shorter compared to the previous course edition 2018-2022). Additional comments about the method thesis are that there should be a checklist of what is needed to be included in terms of tables, graphs and robustness checks (there is a checklist already, but this can be improved) and that too much time is spent on practical technicalities/different tests that was not really introduced in the workshops. One response concern that students, when comparing with each other, perceive they receive different inputs from the teachers, and that we should streamline the demand on the method thesis. One response regards that the start of the work with the method thesis came too late for students who wanted to collect their own data, but in the same response, it is also acknowledged that this later deadline suited many of those who wanted to use difference-in-difference or synthetic control method. Finally, one response state that the final presentations felt too hasty and that more time for the following q&a is needed.

We will continue to fine tune the method thesis instructions, and to further align our expectations on the method thesis.

Responses categorized as miscellaneous are that the week before Christmas was stressful and challenging. Students both needed to coordinate their work with assignment 6 while people were leaving for Christmas and start to prepare the work for the Method thesis. Finally, one response state that the course should be placed during the first year of MAES instead of second year. While we better can prepare students for what work they need to do the last week before Christmas, we cannot decide on the sequence of the course on specific programs.